

Assignment 2

- **SESSION 2**

1. List 3 key responsibilities each for a Tester and a Test Leader in a software project, and give one example of how each role contributes during a release cycle.

ANS :

Project Roles: Tester vs. Test Leader

Tester

Key Responsibilities:

1. Test Case Design: Creating detailed, step-by-step test cases and test data based on requirements.
2. Test Execution: Running manual or automated tests and comparing actual results with expected outcomes.
3. Defect Reporting: Logging clear, reproducible bug reports in tracking tools (like Jira) and verifying fixes.

Release Cycle Contribution: During a release cycle, a tester executes regression suites on the final release candidate build to ensure that new code changes haven't broken existing features (e.g., verifying that a new payment gateway update didn't break the existing log-in system).

Test Leader

Key Responsibilities:

1. Test Planning: Defining the overall test strategy, scope, resource allocation, and timelines for the project.
2. Monitoring & Control: Tracking testing progress against metrics, managing blockages, and adjusting schedules as needed.

3. Stakeholder Communication: Reporting test coverage, quality metrics, and exit-criteria status to project managers and clients.

Release Cycle Contribution: At the end of a release cycle, the Test Leader reviews the defect log and test summary report to formally sign off on the release, confirming that all critical bugs are resolved and the software is stable enough for production.

2. Pick any app you use daily (Instagram, Zomato, Paytm, etc.) and write a one-paragraph comparison of how Quality Assurance (QA) and Quality Control (QC) would be performed for that app.

Ans :

QA vs. QC: A Case Study on Instagram :-

For an app like Instagram, Quality Assurance (QA) focuses on preventing defects by optimizing the processes used to build the app before the code is even written. This involves QA engineers collaborating with product managers to review requirements for a new feature (like "Instagram Reels"), establishing coding standards, and setting up automated CI/CD pipelines to ensure code health. Conversely, Quality Control (QC) is a product-oriented, post-coding activity focused on identifying defects. In QC, testers actually open a beta version of the Instagram app on various iOS and Android devices to execute test cases—such as validating that adding a music sticker doesn't crash the video, or checking if the scroll feed lag complies with performance benchmarks.

3. Explain the difference between Testing and Debugging by describing what a tester does vs what a developer does when a bug is found in a Flipkart-style shopping cart feature.

Ans :

Testing vs. Debugging: The Shopping Cart Scenario

Imagine a bug in a Flipkart-style app: When a user adds two identical items to the cart, the total price doubles correctly, but applying a 10% discount coupon deducts 10% from only one item instead of the total cart value.

What the Tester Does (Testing): The tester discovers the issue during test execution. Their job is to identify, isolate, and report the bug. They will document the exact steps to reproduce it (e.g., 1. Add two 5000 Rs.shoes to the cart. 2. Apply coupon "SAVE10". 3. Observe total is 950 Rs. instead of Rs. 90). They take screenshots or logs, create a ticket in Jira, assign it a severity level, and pass it over to the development team. The tester does not look at the source code to fix it.

What the Developer Does (Debugging): Once the developer receives the ticket, they perform debugging to find the root cause and fix the issue. They will run the application locally, attach a debugger, and step through the backend cart-calculation code. They might find that the loop calculating the discount is terminating prematurely or that the discount function is operating on `item_price` instead of `total_cart_price`. Once the lines of faulty code are identified, the developer rewrites the logic, tests it locally to ensure it works, and submits a code fix.

4.Create a basic test plan skeleton in a Word or Google Doc for a new feature in a food delivery app (like Swiggy) — include at least these sections: Test Plan ID, Feature Description, Test Scope, Roles & Responsibilities, Test Environment, and Test Deliverables.

ANS :

1.Test Plan ID : TP- FDA - 01

2. Feature Description

This test plan covers the new "Live Order Tracking" feature. This enhancement introduces a real-time Google Maps interface for users after checkout. It must accurately reflect four distinct delivery milestones: Order Accepted, Food Being Prepared, Valet En Route to Restaurant, and Valet Delivering to Doorstep. It also includes a live ETA counter and an in-app "Call Driver" masking toggle.

3. Test Scope

In-Scope:

- Functional testing of map rendering, GPS icon movement, and status transitions on iOS and Android.
- Negative testing (e.g., how the map behaves if the user loses internet connection mid-delivery).
- Integration testing between the Driver App status updates and the Customer App interface.

Out-of-Scope:

- Backend database migration performance (handled by the Core Infra team).
- Third-party Google Maps API server outages.

4. Roles & Responsibilities

- Test Lead (1): Responsible for creating this plan, assigning modules to testers, monitoring daily bug burn-down charts, and signing off on the release.
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- QA Engineers / Testers (2): Responsible for writing specific test scripts, executing manual tracking simulations, setting up mock GPS coordinates, and logging defects.
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- DevOps Engineer (1): Responsible for deploying test builds to the staging environment and ensuring mock driver APIs are active.

5. Test Environment

- Handsets & OS: iPhone 14/15 (iOS 17 & 18), Samsung Galaxy S23 (Android 14).
- Network Conditions: Tested across 4G, 5G, and simulated patchy networks (using network throttling tools to mimic elevators/basements).
- Software Build: Staging Environment Build v4.12.0-RC1 connected to the sandbox Google Maps API gateway.

6. Test Deliverables

- Comprehensive Test Case Suite (stored in TestRail).
- Daily Execution & Status Summary Reports.
- Defect Log Matrix (Jira Dashboard link).
- Final Sign-off Certificate with remaining open minor anomalies (if any).